Unit 7 Practice Problems

Area, Surface Area, and Volume Practice

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Aligned to the Honors Geometry unit outline provided by the user.

Unit 7 Practice Problems

Directions

Leave answers in exact form with π unless a decimal is requested. Work all problems first, then check the answer bank and detailed solutions.

Problems

1. Find the area of a parallelogram with base 12 and height 7.

2. Find the area of a triangle with base 18 and height 11.

3. Find the area of a trapezoid with bases 10 and 16 and height 7.

4. Find the area of a rhombus with diagonals 14 and 20.

5. Find the exact area of a regular hexagon with side length 6.

6. Find the area of a circle with radius 9.

7. Find the circumference of a circle with diameter 14.

8. Find the area of a sector with radius 12 and central angle $60°$.

9. Find the arc length for radius 10 and central angle $72°$.

10. Find the area of $\triangle ABC$ with $A(0,0)$, $B(8,0)$, $C(2,5)$.

11. Find the area of the polygon with vertices $(0,0), (6,0), (6,2), (2,5), (0,3)$.

12. Find the surface area of a rectangular prism with dimensions 4, 5, and 6.

13. Find the volume of a rectangular prism with dimensions 4, 5, and 6.

14. Find the volume of a cylinder with radius 3 and height 10.

15. Find the surface area of a cylinder with radius 4 and height 9.

16. Find the volume of a cone with radius 6 and height 8.

17. Find the volume of a pyramid with base area 49 and height 12.

18. Find the volume of a sphere with radius 5.

19. An object has mass 450 g and volume 30 cm^3. Find its density.

20. Water in a cylinder of radius 5 rises from height 10 to 13. Find the displacement volume.

21. State Cavalieri's Principle in one sentence.

22. Model a tree trunk as a cylinder with radius 1.5 ft and height 20 ft. Find its volume.

23. Find the surface area of an open-top rectangular box with length 10, width 8, and height 5.

24. If a solid is dilated by scale factor 2, by what factor does volume change?

25. Find the perimeter of rectangle $A(0,0)$, $B(7,0)$, $C(7,4)$, $D(0,4)$.

Answer Bank

1. 84.

2. 99.

3. 91.

4. 140.

5. $54\sqrt{3}$.

6. 81π.

7. 14π.

8. 24π.

9. 4π.

10. 20.

11. 22.

12. 148.

13. 120.

14. 90π.

15. 104π.

16. 96π.

17. 196.

18. $\frac{500}{3}\pi$.

19. 15 g/cm^3.

20. 75π.

21. Same height and equal cross-sectional areas at every height imply equal volumes.

22. 45π ft^3.

23. 260.

24. 8.

25. 22.

Detailed Solutions

1. Parallelogram area is $A = bh = 12 \cdot 7 = 84$.

2. Triangle area is $A = \frac{1}{2}bh = \frac{1}{2}(18)(11) = 99$.

3. Trapezoid area is $A = \frac{1}{2}(b_1 + b_2)h = \frac{1}{2}(10 + 16)(7) = 91$.

4. Rhombus area using diagonals is $A = \frac{1}{2}d_1 d_2 = \frac{1}{2}(14)(20) = 140$.

5. A regular hexagon is six equilateral triangles. Formula is $A = \frac{3\sqrt{3}}{2}s^2$. With $s = 6$, $A = \frac{3\sqrt{3}}{2}(36) = 54\sqrt{3}$.

6. Circle area is $A = \pi r^2 = \pi(9)^2 = 81\pi$.

7. Circumference is $C = \pi d = 14\pi$.

8. Sector area is $\frac{60}{360}\pi(12)^2 = \frac{1}{6}(144\pi) = 24\pi$.

9. Arc length is $\frac{72}{360} \cdot 2\pi(10) = \frac{1}{5}(20\pi) = 4\pi$.

10. Use base $AB = 8$ and height 5 from C to the x-axis. Area is $\frac{1}{2}(8)(5) = 20$.

11. Use the shoelace formula. The forward sum is $0+12+30+6+0 = 48$. The backward sum is $0+0+4+0+0 = 4$. Area is $\frac{1}{2}|48 - 4| = 22$.

12. Surface area is $2(lw + lh + wh) = 2(4 \cdot 5 + 4 \cdot 6 + 5 \cdot 6) = 2(20 + 24 + 30) = 148$.

13. Volume of a rectangular prism is $lwh = 4 \cdot 5 \cdot 6 = 120$.

14. Cylinder volume is $V = \pi r^2 h = \pi(3)^2(10) = 90\pi$.

15. Cylinder surface area is $2\pi r^2 + 2\pi rh = 2\pi(4)^2 + 2\pi(4)(9) = 32\pi + 72\pi = 104\pi$.

16. Cone volume is $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(6)^2(8) = 96\pi$.

17. Pyramid volume is $V = \frac{1}{3}Bh = \frac{1}{3}(49)(12) = 196$.

18. Sphere volume is $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(5)^3 = \frac{500}{3}\pi$.

19. Density is mass divided by volume: $450/30 = 15$ g/cm^3.

20. The water rises by 3. Displacement volume is $\pi r^2 \Delta h = \pi(5)^2(3) = 75\pi$.

21. Cavalieri's Principle says that two solids with the same height and equal cross-sectional area at every height have equal volumes.

22. Cylinder volume is $V = \pi(1.5)^2(20) = 45\pi$ ft^3.

23. An open-top box has base plus four sides: base $10 \cdot 8 = 80$, two long sides $2(10 \cdot 5) = 100$, and two short sides $2(8 \cdot 5) = 80$. Total 260.

24. Volume scales by the cube of the length scale factor. With scale factor 2, volume factor is $2^3 = 8$.

25. The side lengths are 7 and 4. Perimeter is $2(7 + 4) = 22$.